

Yann Chaubet

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RESEARCH INTERESTS

Hyperbolic dynamics, transfer operators, Pollicott–Ruelle resonances, microlocal analysis, dynamical zeta functions, spectral geometry, geometric topology.

POSITIONS

- May 2026 : Visiting researcher, University of Melbourne.
- 2023– : Assistant Professor (maître de conférences), Nantes Université.
- 2022–2023 : Herschel Smith Research Fellow, University of Cambridge (By-Fellow of Churchill College).

EDUCATION

- 2019–2022 : PhD in Mathematics, Université Paris-Saclay. Advisor: Colin Guillarmou.
On some geometrical applications of the spectral theory of hyperbolic flows.
- 2014–2019 : École normale supérieure, Paris.

PREPRINTS

- Geodesic Lévy flights on Zoll surfaces (with E. J. Godfried and L. Tzou), arXiv:2606.28744.
- Minimax spectral estimation of weighted Laplace operators (with V. Divol), arXiv:2511.22694.
- Nonlinear chaotic Vlasov equations (with D. Han-Kwan and G. Rivière), arXiv:2407.04426.

PUBLICATIONS

- Combinatorial zeta functions counting triangles (with L. Bénard, N. V. Dang and T. Schick), *Journal of Topology* **19** (2026), e70075.
- Geodesic Lévy flights and expected stopping time for random searches (with Y. G. Bonthonneau, T. Lefeuvre and L. Tzou), *Probability Theory and Related Fields* **191** (2025), 235–285.
- Dynamical zeta functions for billiards (with V. Petkov), *Annales de l’Institut Fourier*, Tome **76** (2026), no. 4, pp. 1537–1592.
- Resolvent of vector fields and Lefschetz numbers (with Y. G. Bonthonneau), *Pure and Applied Analysis* **6** (2024), no. 2, 521–540.
- Dynamical torsion for contact Anosov flows (with N. V. Dang), *Analysis & PDE* **17** (2024), no. 8, 2619–2681.
- Closed geodesics with prescribed intersection numbers, *Geometry & Topology* **28** (2024), no. 2, 701–758.
- Closed billiard trajectories with prescribed bounces, *Annales Henri Poincaré* **24** (2023), no. 3, 1051–1075.
- Poincaré series for surfaces with boundary, *Nonlinearity* **35** (2022), no. 12, 5993–6013.

Book chapter.

- Dynamical zeta functions. Chapter 10 in *Microlocal Analysis in Hyperbolic Dynamics and Geometry*, by T. Lefeuvre, *Cours Spécialisés de la SMF*, vol. **32** (2026), pp. 219–264.

AWARDS AND GRANTS

- 2026 – ANR JCJC grant, 200 k€.
- 2025 – International Emerging Action (partnership with Melbourne University), 28 k€.
- 2025 – PEPS grant, 4.4 k€.
- 2023 – PULSAR grant, 10 k€.
- 2022 – Herschel Smith Research Fellowship, 15 k€/year.

INVITED TALKS

Mini-courses.

- October 2025, Journée Rennes–Nantes d’analyse, Rennes (2 lectures).
- December 2025, Conférence Géométrie et Dynamique, Aussois (3 lectures).

International conferences.

- Three Days Workshop in Dynamics, Geometry and Analysis, Melbourne, 2026.
- Rigidity Phenomena in Dynamics and Spectral Theory, Zurich, 2024.
- Hyperbolic Dynamics and Semiclassical Analysis, Paris, 2023.
- Analytic Techniques in Dynamics and Geometry, Les Diablerets, 2023.
- Rencontre Aléatoire, Dynamique et Spectre, Grenoble, 2023.
- École de Printemps Géométrie et Dynamique, Lille, 2023.
- Paris–London Analysis Seminar, London, 2023.
- Geometrical Inverse Problems, Linz, 2022.
- Rigidity Problems in Geometry, Roscoff, 2022.
- Spectra and Dynamics on (Locally) Symmetric Spaces, Paderborn, 2022.

Seminars. University of Queensland (2026), Monash University (2026), Pisa (2025), Brest (2025), Tours (2024), Quimper (2024), Nantes (2024), Warwick (2024), Bristol (2023), London (2023), Cambridge (2023), Marseille (2022), IMJ-PRG (2022), Montpellier (2022), Bochum (2022), Pisa (2022), Bordeaux (2022), Orsay (2020–2021), Marseille (2020).

CONFERENCE ORGANIZATION

- Nov 2026 – Journées quimpéroises des jeunes chercheuses en analyse microlocale, Quimper.
- Sep 2026 – Workshop Semiclassical analysis along the Loire, Nantes.
- Nov 2025 – Journées brestoises des jeunes chercheuses et chercheurs en analyse microlocale, Brest.
- Sep 2025 – Three Days Workshop on Geometry, Analysis and Dynamics, Longeville-sur-Mer.
- Oct 2024 – ANR Adyct Workshop, Nantes.
- May 2024 – Workshop on Pseudo-Holomorphic Curves, Teyran.